

Research article

Investigating response behavior through TF-IDF and Word2vec text analysis: A case study of PISA 2012 problem-solving process data

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ABSTRACT

The process data in computer-based problem-solving evaluation is rich in valuable implicit information. However, its diverse and irregular structure poses challenges for effective feature extraction, leading to varying degrees of information loss in existing methods. Process-response behavior exhibits similarities to textual data in terms of the key units and contextual relationships. Despite the scarcity of relevant research, exploring text analysis methods for feature recognition in process data is significant. This study investigated the efficacy of Term Frequency-Inverse Document Frequency (TF-IDF) and Word to Vector (Word2vec) in extracting response behavior features and compared the predictive, analytical, and clustering effects of classical machine learning methods (supervised and unsupervised) on response behavior. An analysis of the PISA 2012 computer-based problem-solving dataset revealed that TF-IDF effectively extracted key response behaviors, whereas Word2vec captured effective features from sequenced response behaviors. In addition, in supervised machine learning using both methods, the random forest model based on TF-IDF performed the best, followed by the SVM model based on Word2vec. Word2vec-based models outperformed TF-IDF-based ones in the F1-score, accuracy, and recall (except for precision) across the logistic regression, k-nearest neighbor, and support vector machine algorithms. In unsupervised machine learning, the k-means algorithm effectively clustered different response behavior patterns extracted by these methods. The findings underscore the theoretical and methodological transferability of these text analysis methods in educational and psychological assessment contexts. This study offers valuable insights for research and practice in similar domains by yielding rich feature representations, supplementing fine-grained assessment evidence, fostering personalized learning, and introducing novel insights for educational assessment.

1. Introduction

Although the abundant cognitive information and strategies inherent in problem-solving processes hold significant value [1,2], their extraction poses challenges in traditional data analysis methods [3]. Computer-based assessments, as highlighted by international assessments such as the Programme for International Student Assessment (PISA) and Trends in International Mathematics and Science

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Study (TIMSS) [1,4,5], enable the recording of behavior sequences and response processes. These sequences are rich in implicit information, including strategies, thinking processes, and metacognition [6], and marked by complexity, high noise, and irregularity [7, 8], making them ill-suited for conventional analysis techniques. Converting unstructured behavior sequences into structured numerical data through feature extraction is critical for the accuracy and reliability of model analysis. However, traditional methods fall short in this regard, thereby necessitating the exploration of new, effective feature extraction techniques.

Despite ongoing research into feature extraction methods in educational assessment, current technologies face challenges and opportunities in extracting and generalizing behavioral information [9]. The main challenges encountered in our investigation stem from the intricate and diverse nature of the data, as well as the demand for precise analytical methods. Process data not only capture the final responses of students but also document their entire problem-solving process, including tool usage, temporal sequences, and interactive choices [10]. In addition, process-based responses typically manifest as sequences, exhibiting significant variations in length, structure, and complexity. For example, when tackling mathematical problems, each student may employ distinct strategies, resulting in differing numbers of steps. This complexity and irregularity necessitate the use of advanced data analysis approaches, such as machine learning, deep learning, network analysis, and text analysis, to standardize and extract pertinent feature information [9, 11].

Data structure similarities enable interdisciplinary application of analytical techniques. Process-response data shares many similarities with textual data regarding attribute characteristics. Firstly, both exhibit a serialized nature, being collections of elements (such as words, sentences, or user actions) arranged in a specific sequence, thus imbuing significance not just to individual elements but also to their order and combination. Secondly, both types of data are unstructured, lacking fixed patterns or formats. Textual data may vary in sentence lengths and structures, while process-response data can include various types of user interactions and operations, making them difficult to represent using conventional structured database tables. Thirdly, both types of data are multidimensional, encompassing various levels or dimensions of information. Textual data conveys not only surface-level text but also sentiment, semantic structure, and context. Similarly, process-response data can reflect strategy choices, problem-solving processes, and cognitive patterns, beyond recording operation sequences. Fourthly, the interpretation of both types of data often depends on contextual cues. Textual data may be influenced by surrounding text or broader topics, while understanding process-response data can require considering the specific task or goal of user operations. These similarities facilitate interdisciplinary exchange and methodological integration to address the analytical demands of process data.

The impact of text analysis methods on feature extraction are worthy of in-depth exploration [9,12]. The effectiveness of such methods in extracting behavior characteristics within complex problem-solving contexts remains uncertain. Moreover, given the diverse attributes of these methods, determining the most suitable approach for extracting behavior features with sequential relations remains a topic of ongoing exploration. This study employed two text feature extraction methods—Term Frequency-Inverse Document Frequency (TF-IDF) and Word to Vector (Word2vec)—to advance research in behavioral sequence feature extraction and achieve more precise extraction of procedural cognitive information during problem-solving. These techniques were applied to analyze the effects of feature extraction based on answer-behavior sequences in PISA 2012 [6]. Additionally, supervised machine learning methods were employed to predict and elucidate the answering performance of respondents, while an unsupervised machine learning method was used to cluster behavior sequences, thereby facilitating the differentiation and analysis of respondent groups with distinct behavioral patterns.

2. Literature review

This section first provides an overview of the current state of computer-based assessment for evaluating problem-solving abilities and identifies the need for advancements in this field. Second, it introduces the background of analyzing process-oriented response behavior sequences generated by computer-based assessment, including process data, behavior feature extraction, text-based methods, and statistical models, is introduced.

2.1. Computer-based assessment of problem-solving competency

Computer-based assessments have been applied to evaluate higher-order cognitive abilities, including problem-solving [6,13–16], providing both theoretical and practical evidence. Firstly, realistic and complex problem situations can be presented [17]. Secondly, the process data collected via a computer-based assessment can provide an effective implicit judgment basis for the cognitive process in problem-solving that cannot be measured via conventional assessments [18,19]. This enhances the validity of the interpretation of assessment scores [20] and provide a reliable basis for improvement in the accurate diagnosis of learning and the enhancement of learning development [1,21]. The Organization for Economic Co-operation and Development (OECD) conducted computer-based assessments and studies on process behaviors focusing on problem-solving and collaborative problem-solving skills in 2012, 2015, and 2022 [4–6]. Other large-scale international assessments have also placed significant emphasis on these areas [1,22].

2.2. Behavioral sequence data analysis

Sequential data from process-based answering behaviors exhibit complexity and interrelatedness, challenging traditional statistical analysis methods. With recent technological advancements, researchers are exploring novel approaches for feature extraction. Text analysis methods have gained attention as interdisciplinary avenues for analysis. Given their potential in capturing attributes relating to key behaviors and action correlations within behavioral sequences, further exploration and validation of these text methods for

feature extraction are justified.

2.2.1. Process data

Process data can reflect information besides the total score [8,23]. In the problem-solving process of human-computer interaction, the process data is in the form of a series of log files with time, action, and other attributes, such as the respondent's behavioral pattern, speed, answering strategy, solution thinking, answer patience, and even learning trajectory [24,25]. Compared to time and strategy indicators, behavior sequences are less dependent on the specific topic and a researcher's prior educational domain knowledge. Feature extraction research on behavioral sequences has better scalability and can be extended to other topics, thereby assisting researchers in capturing evidence of cognitive processes from respondents from a data-driven perspective.

However, owing to the structural irregularity of behavioral sequences, their unstructured nature poses several challenges to conventional evaluation and analysis methods. Unstructured data refers to data whose structure is irregular, i.e., there is no predefined data analysis model, and the data cannot be represented using a two-dimensional logical table of the database [8], and a matrix cannot be easily generated [26]. It is difficult to extract meaningful results from the large amount of information generated during the interaction of respondents [3]. As a result, existing measurement models and standard statistical methods cannot be directly applied. The reliable extraction of potential information from behavior sequences and the conversion of the sequences into structured data is an active area of research.

2.2.2. Behavior sequence feature extraction

Feature extraction is crucial in analyzing unstructured behavior sequences. It involves extracting potential information using data-driven methods, converting the sequences into structured data that computers can understand. This process greatly impacts the accuracy and reliability of subsequent model analysis. A review of the relevant literature has shown that response behavior sequence feature extraction methods include.

- 1) Frequency statistics: Counts the occurrences of different behaviors in a sequence to realize the numerical representation of the behavior sequence data. Many studies on the response behavior sequence utilize this method for feature extraction of the behavior sequences [27–29].
- 2) N-gram: This method is derived from text feature extraction, which refers to the occurrences of N consecutive words in a text. In the sequence of answer-behaviors, it refers to the number of times N continuous behaviors appear in a sequence, which is a derivative of the frequency statistics method. Analysis of behavioral sequences via N-gram methods has been shown to successfully identify strategy changes [30].
- 3) Longest common subsequence: This involves finding the common longest subsequence in a sequence set of two sequences. For the behavior sequence, this involves finding the longest subsequence that is common between each answer-behavior sequence and the optimal answer-behavior sequence, and calculating the distance and similarity with the optimal answer-behavior sequence on this basis. The longest common subsequence can be used to establish valuable strategies by using the distance and similarity between different response sequences [31].
- 4) Multi-dimensional scaling: This method builds features based on relative differences, and locates them in vector space so that similar elements are closer and different elements are further apart. If the difference between two behavior sequences is appropriately summarized by the difference measure, the coordinates obtained at the multi-dimensional scale can be regarded as the characteristics of storing the original process information. Interpretable latent variables can be extracted from different similarity matrices of the answering process via multi-dimensional scaling transformation [32].
- 5) Autoencoder: This method is an unsupervised artificial neural network that compresses the original input data and reshapes the input features. The autoencoder can compress the response sequence information matrix into a standard numerical vector, from which valuable information can be extracted [26].

The aforementioned methods have their advantages and disadvantages in terms of feature extraction of behavior sequences. Frequency statistics focus on the occurrence count of behaviors in a sequence, which is easy to use and understand. However, sequential behavior order might be overlooked, limiting data comprehension depth. Additionally, complex inter-behavior relationships or patterns may not be captured by relying solely on behavior occurrence counts, resulting in incomplete information extraction. The N-gram method, which is based on frequency statistics, considers behavior sequences but exhibits an increased feature dimensionality with a larger N, leading to greater model complexity. Moreover, it may inadequately capture the long-term dependencies or behavior sequence complexity, as illustrated in climate control data with over 5000 bi-gram behaviors corresponding to 128 unigram behaviors. The longest common subsequence method can capture information beyond temporal and action frequency aspects. However, its effectiveness relies on the presence of an optimal or benchmark answer sequence for comparison, which may not always be feasible or clearly defined across all contexts. Furthermore, its scalability is limited and it may not be suitable for all types of problems. Although multidimensional scaling is effective in identifying data structures or clusters, it may pose interpretability challenges as the resulting dimensions or latent variables may not directly correspond to intuitive or easily understandable features of behavioral sequences. Similarly, autoencoders, adept at handling complex, high-dimensional data, encounter significant interpretability issues with their compressed features. The design of the transformation process is not intended for easy human comprehension and the quality of the extracted features is heavily reliant on the architecture and training of the autoencoder, thus requiring careful tuning and validation [26].

It is evident that achieving effective analysis of process data is strongly dependent on matching the structural characteristics of the

data with the attributes of the methods used. As discussed previously, process data share structural similarities with textual data in terms of the sequential nature, unstructured format, multidimensionality, and contextual dependence. These similarities make it feasible to apply text analysis methods to the analysis of process-oriented response data. Furthermore, the complexity and volume of elements in text data provide favorable conditions for text analysis methods to handle process-oriented response data effectively and mitigate the limitations of the aforementioned methods. However, there remains a significant lack of exploration and experimentation in this area, underscoring the need for further investigation.

2.2.3. Text analysis and behavior sequence feature extraction

In the behavior sequence, the recognition of operation features is similar to the recognition of text features, and more evidence is needed. Unlike data mining which deals with well-structured data, text data mining deals with unstructured documents. Thus, one of the objectives of text data mining research is to convert text into numeric vectors, i.e., the numerical representation of text [33]. The logic of this transformation is very close to the requirement of transforming unstructured response data into structured features for extraction. Although text feature extraction methods have been applied in numerous fields to numerically represent text-like sequence data to achieve superior results [34–36], thus far, few studies have applied text methods to feature extraction in behavioral sequences, and the implemented methods (such as N-gram) are limited by the feature dimension. In addition, the order information that can be reflected is limited [30]. To investigate more effective feature recognition methods and to exploit the application value of text analysis methods in sequence analysis, it is worth exploring and verifying the recognition effects of other text methods in this process.

Text analysis methods need to capture key behaviors and explore the sequential relationships between them adeptly for more effective feature extraction from behavioral sequences. Different text analysis methods place distinct emphases on attribute characteristics. For instance, FastText emphasizes subword decomposition, making it suitable for handling variations within words, such as plurals, tenses, and other inflections. Bag-of-words (BOW) represents text with word vectors, but it only considers word frequency and cannot capture the sequential or semantic relationships between words. GloVe optimizes word vectors using global co-occurrence information to capture both the local and global semantics. It excels on large-scale corpora, effectively handling rare words and long-tail distributions. N-grams capture the relationships in text based on consecutive sequences of n words or characters and have been used in similar research [10], [11]. However, they pose the risk of dimensionality explosion in the feature space. Thus, feature selection or dimensionality reduction is necessary and warrants further exploration. In comparison, TF-IDF and Word2vec are suitable for capturing the feature extraction needs of analyzing process data from answering behaviors, specifically in terms of capturing key behaviors and behavioral relationships. Therefore, they are worth considering for feature extraction in this context.

TF-IDF has shown promising potential in extracting key features relating to response behaviors. As a typical text analysis method, TF-IDF evolved from IDF (Inverse Document Frequency), and is a derivative method of frequency statistics [37]. The basic idea of TF-IDF originates from language modeling theory, where words in documents are divided into two categories: keywords and non-keywords [38]. The keywords are closely related to the topic of the document, and the criticality of the words can be evaluated using TF-IDF. TF-IDF compensates for the shortcomings of the frequency statistical method and focuses on the value of key behaviors. The TF-IDF algorithm with a criticality attribute for evaluating keywords has been used to automatically score essays [39], for curriculum model optimization [40] and test question classification [31]. In the complicated sequence of answer-behaviors, judgments are made on key and non-critical behaviors, analogous to textual data. However, common behaviors that appear repeatedly in many sequences are not necessarily a good key discriminant behavior and should be given a lower weight. However, research utilizing this method to identify key behaviors is scarce. This operational principle is worth exploring in the field of psychological and educational assessment.

Word2vec word embeddings offer the potential to facilitate the nuanced contextual analysis of process-oriented response data by extracting the semantic relationships among words. Based on neural networks, Word2vec have greatly improved the efficacy of text feature extraction [41,42] and has been extended to other domains beyond textual data [34,43]. The network method is trained and maps words to a low-dimensional vector space, capturing the semantic relationship between them, so that semantically similar words are close in space, and word vectors are used to represent the semantic information of words. Among these approaches, the Word2vec method excels at various textual tasks [41]. As a word embedding method, the Word2vec model represents each word using the hidden layer weight after training, mapping each word to a vector to represent the relationship between words. It should be emphasized that Word2vec extends its underlying principles to action sequences through the vectorization of words, thereby enabling the vectorization of actions. According to the similarity in the structure of the behavior sequence and the text data, there may be potential "context" relationships between actions in action sequences. Thus, Word2vec can be extended to the feature extraction of behavior sequences. This logic has also been applied and evaluated in related research [36].

As mentioned previously, the principles of Word2vec can be applied to action sequences to vectorize actions. Given the similarity in the structure of the behavior sequence and the text data, there may be potential "context" relationships between actions in action sequences. Thus, Word2vec can be extended to the feature extraction in behavior sequences. This logic has also been applied and evaluated in related research [36]. Behaviors can be mapped to a low-dimensional vector space to capture potential relationships via the "context" of the behaviors, forming the basis for the feature extraction of Word2vec in behavior sequence data. This transferable application has not been investigated in the study of problem-solving behavior sequences and is worth exploring and verifying.

However, research on extracting procedural answer-behavior features in the fields of education and psychological assessment using these two methods remains scarce. Although the advantages of Word2vec and TF-IDF have been progressively validated in education and psychology, such as patterns of language use on social media [44–46] and insights into user psychology and behavior [47,48], most of these validations were based on text material analysis. Research focusing on log file analysis behavior, such as examining the interactive behaviors of students on online learning platforms to identify learning patterns [49], remains relatively rare. A review of

the existing literature indicates that employing these two approaches to study student response behaviors within educational and psychological assessment is very uncommon.

Thus, further research is needed to explore the value of TF-IDF and Word2vec in the feature extraction and analysis of process data from answering behaviors. Considering the theoretical characteristics of text feature extraction methods, their wide application in related fields, and the similarity between the structure of behavior sequences and text data, TF-IDF can identify keywords, thereby aiding in understanding crucial content and facilitating classification or clustering analysis of student behaviors. Meanwhile, Word2vec captures semantic relationships, assisting in recognizing similar patterns or error modes in answering behaviors. Therefore, both methods offer valuable insights for analyzing process data.

2.2.4. Statistical modeling

After effectively extracting features from behavior sequences, subsequent statistical model analysis is necessary to test feature extraction and validity. Compared with conventional measurement methods for analyzing process data, machine learning is more suitable for behavioral sequence data with complex structure and high noise, and has demonstrated effective analysis results [50]. In educational evaluation, machine learning methods can be used to analyze the learning and interaction behaviors of students and to establish student models [51].

Machine learning have been proven to be effective in process data analysis, but not based on features extracted from text analysis. Supervised learning methods used for process data analysis include neural networks, decision trees, random forests, support vector machines (SVM), Naive Bayes, and k-nearest neighbor (KNN) [52,53]. For unsupervised learning, cluster analysis is one of its main applications, with the k-means algorithm being one of the most widely used [53,54]. However, there are few studies on feature extraction based on text analysis for problem-solving process data, and both supervised and unsupervised machine learning are still used to model analysis.

2.3. Research questions

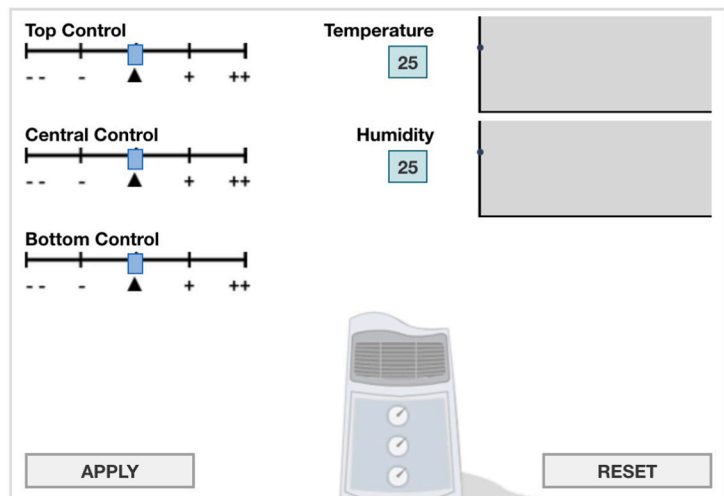
This study aimed to evaluate the feasibility of TF-IDF and Word2vec for analyzing response behavior sequence, to extract valuable information from the response behavior sequence for problem-solving, to utilize machine learning models to predict the response

CLIMATE CONTROL

You have no instructions for your new air conditioner. You need to work out how to use it.

You can change the top, central and bottom controls on the left by using the sliders (→). The initial setting for each control is indicated by ▲.

By clicking APPLY, you will see any changes in the temperature and humidity of the room in the temperature and humidity graphs. The box to the left of each graph shows the current level of temperature or humidity.



Question : CLIMATE CONTROL

Find whether each control influences temperature and humidity by changing the sliders. You can start again by clicking RESET.

Draw lines in the diagram on the right to show what each control influences.

To draw a line, click on a control and then click on either Temperature or Humidity. You can remove any line by clicking on it.

SUBMIT

RESULTS

Top Control

Temperature

Central Control

Humidity

Bottom Control

Fig. 1. PISA 2012 Problem-Solving item: Climate control.

performance of respondents, and to further analyze the behaviors and strategies required for responses. Groups with different response behaviors and strategies will then be distinguished and analyzed. This helps to identify different types of respondents more accurately and facilitates subsequent model analysis. Moreover, the predictions hold theoretical value and practical significance in the identification and accuracy of procedural response behavior sequences. Therefore, the following research questions are proposed.

- 1) Can TF-IDF and Word2vec effectively extract key behaviors and behavior sequence features from problem-solving process-response data?
- 2) Can supervised machine learning models, utilizing behavior features extracted by TF-IDF and Word2vec, accurately predict respondents' performance and strategies?
- 3) Can unsupervised k-means clustering analysis, using behavior features derived from TF-IDF and Word2vec, effectively differentiate behavior patterns and success rates?

3. Materials and design

The dataset, research design, and analysis methods employed in the study are outlined in the following subsections.

3.1. Data set

The dataset for this study was obtained from the PISA 2012 computer-based assessment of complex problem-solving. The assessment series demonstrated strong reliability and validity [6]. The process data were validated to reflect the internal cognitive processes involved in complex problem-solving, and they are extensively used in research and analysis in the fields of education and psychological assessment [55–58].

The specific action sequences analyzed were derived from the problem-solving task "climate control (CP02501)" within the assessment, as shown in Fig. 1 [6]. Problem scenarios were presented via computer simulations. The left side of the page provided the scenario for problem-solving, in which students were tasked with exploring changes in temperature and humidity within a room by autonomously adjusting certain sliders, without instructions for a new air conditioner. The right side featured an interactive area displaying three sliders—the top, central, and bottom controls—for adjusting positions with values ranging from -2 to 2 , along with the corresponding temperature and humidity change trends. Students observed the changes by adjusting the position of one or more sliders, and finally drew a connecting line to indicate the connection between the input and output. They could click "Apply" to observe the effect of the settings, and observe the temperature and humidity changes, or reset the settings by clicking the 'Reset' button. Clicking the "Draw" button drew lines to answer the connecting questions in the bottom-right corner. All these click operations were meticulously recorded in the form of log files, serving as the raw data for the behavioral sequences during the problem-solving process.

The climate control dataset comprises a total of 16,763 samples. Answers were scored as incorrect (0 points), partially correct (1 point), and correct (2 points). In this study, a binary scoring format (0/1) was used to address the task-specific nature of the response data and the research objectives. Partial completion, which did not lead to the resolution of the problem context, was considered incorrect. Given the complexity of processing procedural response behaviors using text analysis methods, the scoring format was consolidated to investigate the effectiveness of text methods in extracting features of correct response behaviors. In the dataset, 9129 samples (54 %) answered correctly (counted as 1), and 46 % answered incorrectly (counted as 0). This database is publicly available and can be obtained from the following link: <https://www.oecd.org/pisa/data/pisa2012database-downloadabledata.htm>. The climate control dataset originated from the ProcData package in R language [26].

Table 1 shows a sample of behavior sequence data after preprocessing. Each line represents the number and behavior sequence of the respondent. "Start" indicates the initiation of the answering process, "end" indicates the termination of the answering process, "reset" refers to the resetting of the scroll bar, and "draw" describes the operation of drawing connecting lines. Since the study focused on the answering process, the "draw" behavior was not included in the behavior sequence after preprocessing. Other actions recorded in Table 2, such as "1_-1_0", represent a series of actions by a student where the top slider was moved to "+", the middle slider to "-", followed by the clicking of the "Apply" button.

Table 1
Examples of response behavior sequences after preprocessing.

Respondent number	Behavior sequence
ARE000000200039	[start, 0_0_0, 1_2-2, 2_2_2, 2_2_2, 2_2_2, 2_2_2, 2_2-2, 2_2-2, 2_2-2, -2_-2-2, -2_-2-2, -2_-2-2, -2_-2-2, -2_-2-2, -2_-2_0, -2_-2_0, -2_-2_0, -2_0_1, -2_0_1, -2_0_1, -2_0_1, -2_0_1, 0_0_1, 0_0_1, 0_0_1, 0_0_1, 0_0_1, 0_0_1, 0_0_1, 0_0_1, 0_0_1, 0_0_1, 0_0_1, end]
ARE000000200051	[start, reset, -1_0_0, -1_-1_0, -1_-1_-1, -1_0_0, -1_0_0, reset, 2_0_0, reset, 0_2_0, reset, 0_0_2, reset, 0_1_0, reset, 0_-1_0, reset, -1_0_0, reset, end]
ARE000000300079	[start, 1_1_1, reset, 0_0_1, reset, 0_1_0, reset, 1_0_0, end]
ARE000000400093	[start, 1_0_0, 0_0_0, reset, 0_0_0, 0_0_1, 0_0_2, reset, 0_1_0, 0_2_0, -1_2_0, reset, 2_0_0, reset, 2_0_0, 2_0_0, 2_0_0, reset, 0_2_0, 0_2_0, 0_2_0, reset, 0_0_2, 0_0_2, 0_0_2, reset, 1_1_1, end]
ARE000000400117	[start, -2_2-2, -2_-2-2, -1_-1_-1, 2_0_0, 0_2_0, 0_0_2, end]

Table 2
Sample of frequency statistical features.

Respondent number	start	0_0_0	1_2_-2	2_2_2	2_2_-2	2_-2_-2	-2_-2_-2
ARE000000200039	1	1	1	5	2	0	0
ARE000000200051	1	0	0	0	0	0	0
ARE000000300079	1	0	0	0	0	0	0
ARE000000400093	1	2	0	0	0	0	0
ARE000000400117	1	0	0	0	0	0	0

3.2. Research design

The design for research question 1 is depicted in Fig. 2. This part involved extracting behavioral features from process-based response data using the TF-IDF and Word2vec text methods, and comparing them with traditional rating statistics. Subsequently, logistic regression was applied to the extracted features using the different methods to analyze their differences and advantages in behavioral feature extraction. To validate the effectiveness of TF-IDF and Word2vec in feature extraction, the absolute values of the logistic model coefficients were used to generate frequency variables and differences in the TF-IDF model scores to verify the identification of key behaviors. In addition, to assess whether Word2vec successfully captured latent behavioral relationships, the cosine similarity and t-distributed stochastic neighbor embedding (t-SNE) visualization methods were used to investigate the relationships between the behavior vectors generated by Word2vec.

The design for research question 2 is shown in Fig. 3. Based on the behavior features and vectors extracted from TF-IDF and Word2vec, supervised machine learning with the KNN, SVM, and random forest algorithms was performed separately to predict and analyze the response reactions, and the optimal hyperparameters of each model were determined using a grid search. Based on the evaluation index F1-score and 5-fold cross-validation, the performance of the model was compared to that of the logistic regression as the baseline. To determine which behaviors and strategies were required to correctly answer the question, the output of the study was analyzed based on the random forest feature importance of TF-IDF.

The design for research question 3 is shown in Fig. 4. Using the behavior features and vectors extracted from TF-IDF and Word2vec, unsupervised machine learning with k-means clustering was conducted separately to explore the clustering patterns of the extracted behavior features. The parameter K was determined using the elbow rule, and the proportional Z test was used to ascertain if there was a significant difference in the correct answer rate among the K groups of respondents, to explain the behavior sequence characteristics between the groups, and to compare the performance of the k-means algorithm based on TF-IDF and Word2vec.

3.3. Methods and evaluation

The TF-IDF and Word2vec text methods were used to handle feature extraction requirements in complex problem-solving response sequences. Subsequently, both supervised models (KNN, SVM, and random forest) and an unsupervised model (k-means) were employed for evaluation.

3.3.1. TF-IDF

TF-IDF is an optimization method for frequency statistics, which demonstrates superior performance compared to the frequency statistical method under specific conditions. This approach utilizes the IDF as a weight to calculate the TF (Term Frequency) by weight. IDF assumes that the importance of a word relative to a document is inversely proportional to its frequency in all the documents. The equation for determining the TF-IDF is as follows:

$$w_{ij} = tf_{ij} \times \log \left(\frac{N}{df_i + 1} \right) \quad (1)$$

In equation (1), w_{ij} is the numerical representation of the i -th word in the j -th document, tf_{ij} is the frequency of the i -th word in the j -th document, that is, the word frequency TF; $\log \left(\frac{N}{df_i + 1} \right)$ calculates the IDF, where N is the number of documents, and df_i is the number of documents that contain the i -th word.

In this study, w_{ij} is the numerical representation of the i -th behavior in the j -th sequence, tf_{ij} is the frequency of the i -th behavior in the j -th sequence, and N is the number of sequences in the sequence set, and df_i is the number of sequences in the set that contains the i -



Fig. 2. Research design for Q1: Comparison of feature extraction methods.

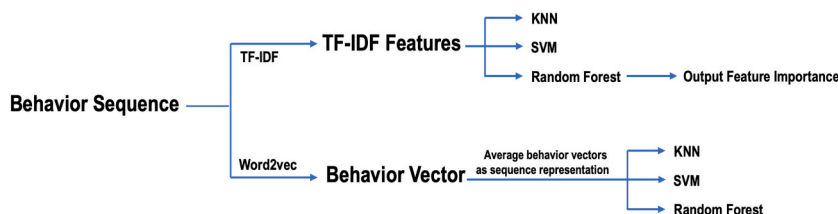


Fig. 3. Research design for Q2: Model comparison with supervised learning.



Fig. 4. Research design for Q3: Model comparison with unsupervised learning.

th behavior.

There are a total of 128 unique behaviors in the climate control behavior sequence dataset. The sample data of the frequency statistics feature is shown in Table 2, and the sample data of the TF-IDF feature is shown in Table 3. In the frequency statistics feature, since the value of each feature is the frequency of the behavior, the value of “start” in each sequence is consistent. However, for the TF-IDF feature, the value of “start” in each sequence is inconsistent. According to equation (1), the TF-IDF variables of “start” and “end” contain the sequence length information, which is inversely proportional to the sequence length.

3.3.2. Word2vec

Word2vec [59] is a neural network probabilistic model for generating word vectors. Its core is a shallow neural network, which uses a sliding window to specify fixed-size context, and attempts to predict the context for the current word or to use the context to predict the current word. The algorithm contains two training models, the CBOW model (Continuous bag-of-words model) and the Skip-gram model (Continuous skip-gram model), both of which are shallow layers composed of an input layer, a hidden layer, and an output layer. The structure of the layer neural network model is shown in Fig. 5.

The CBOW and skip-gram models both use hidden layer weights that are acquired during training to represent word vectors. Although their objectives are similar, they differ in terms of training methods. The CBOW model predicts the current word based on its context, using multiple context words as input to predict the center word. This approach is effective on small datasets and training is quick owing to its use of context word distributional information. Conversely, the skip-gram model predicts the surrounding context based on a given word, using the center word as input to predict its surrounding context words. While skip-gram excels with large datasets and rare words or phrases, its training proceeds at a relatively slower pace compared to CBOW.

In this study, Word2vec was used to convert actions in answer sequences into vectors by treating these actions as words. Given the scale of the dataset and the need to identify patterns within answer sequences, the CBOW model was selected based on several key factors: (1) Small-scale dataset: CBOW is preferred for its rapid convergence on smaller datasets due to its contextual smoothing approach [60]. (2) Computational efficiency: CBOW offers faster training by predicting a single word per sample, which makes it suitable for resource-constrained scenarios. (3) High-frequency word processing: CBOW trains effectively on high-frequency words through context-based predictions. (4) Semantic accuracy: CBOW potentially achieves better semantic precision by considering the entire context, which is particularly important for understanding word meanings. The selection of the CBOW model was driven by the relatively small size of the dataset and the aforementioned considerations.

3.3.3. Statistical modeling methods and evaluation

To validate the efficacy of the two text analysis methods in extracting behavioral features further, several well-established supervised machine learning techniques were used, including KNN, SVM, and random forest, along with the unsupervised machine learning method k-means for prediction and clustering analysis based on the behavioral features. These machine learning

Table 3

Sample of TF-IDF features.

Respondent number	start	0_0	1_2_-2	2_2_2	2_2_-2	2_-2_-2	-2_-2_-2
ARE000000200039	0.02	0.04	0.12	0.30	0.19	0.10	0.39
ARE000000200051	0.06	0.00	0.00	0.00	0.00	0.00	0.00
ARE000000300079	0.16	0.00	0.00	0.00	0.00	0.00	0.00
ARE000000400093	0.05	0.20	0.00	0.00	0.00	0.00	0.00
ARE000000400117	0.12	0.00	0.00	0.00	0.00	0.00	0.45

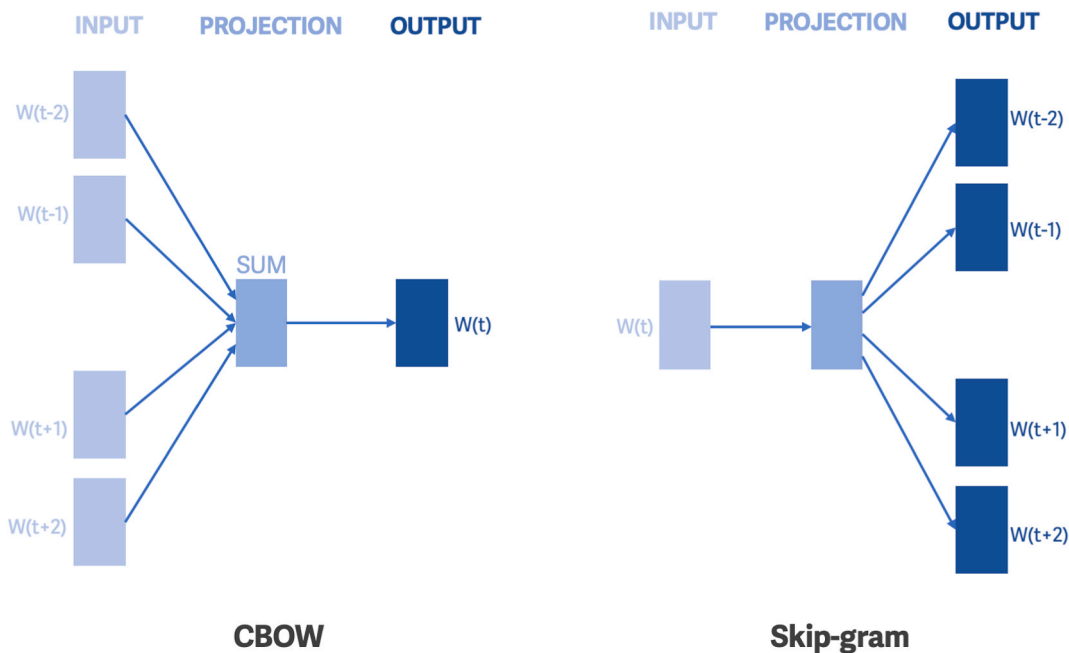


Fig. 5. Structure of the CBOW and Skip-gram models.

methodologies have been applied and validated extensively within the psychology and education domains.

The KNN algorithm, which is recognized for its simplicity, intuitiveness, lack of training requirements, flexibility, and high customizability, is a straightforward yet potent supervised learning method for classification and regression tasks. Its operational principle hinges on the concept of distance or similarity: given a query point (i.e., the data point to be classified or predicted), KNN identifies the "k" closest neighbors within the training dataset and predicts the label or value of the query point based on the information of these neighbors [60]. The KNN algorithm has exhibited significant potential for identifying and predicting patterns in student behavior, learning outcomes, and psychological states within the fields of psychology and education [60–62].

SVM is a powerful supervised learning algorithm that used for both classification and regression tasks. The core principle of SVM is to determine an optimal hyperplane that maximally separates the data points of different classes, making it particularly effective for high-dimensional data classification. SVMs are renowned for their performance in situations where the data dimensionality exceeds the number of samples, as they can maintain high accuracy [63]. SVM can be applied to various tasks, such as predicting academic performance, classifying student behaviors, and recognizing emotional states, in the fields of education and psychometrics. Its ability to handle complex patterns in high-dimensional feature spaces and maintain accuracy with limited samples makes it a valuable tool for these areas [64].

Random Forest is an ensemble learning method that constructs multiple decision trees for classification or regression tasks. It generates different training sets using random sampling and selects features randomly at each node of the decision trees for splitting. The final prediction is made by aggregating the results from multiple decision trees. Random forest is known for its efficient training speed, good generalization ability, and resistance to overfitting [65]. Random forest is used for analyzing and predicting the psychological states and behavioral patterns of individuals in the field of psychometrics. For instance, random forest can aid researchers in identifying factors such as mental health issues and personality traits by examining data from surveys and social media behavior [64].

K-means, which is a prevalent clustering algorithm, partitions data into K clusters based on the proximity to the nearest centroid, performing iterative optimization until criteria such as stable cluster assignments or a set iteration count are met. Its simplicity, efficiency, and scalability make it apt for large datasets [54]. K-means facilitates the analysis of the learning behaviors, performance, and psychological states of students to categorize those with similar traits in educational psychology. This aids in tailoring educational strategies and interventions, allowing for the identification of students at varying learning stages or those requiring extra support [9, 53].

To further evaluate the performance of the model, the commonly used model evaluation method — k-fold cross-validation was utilized to improve the generalization of the model and suppress the influence of randomness on the evaluation. The average value of the K evaluation indicators was used as the final evaluation metric for k-fold cross-validation. The F1-score, accuracy, precision, and recall were employed to assess the effectiveness of the model [66], with the respective formulas provided in equations (2)–(5):

$$F1 - score = \frac{2}{recall^{-1} + precision^{-1}} = 2 \frac{precision \times recall}{precision + recall} \quad (2)$$

$$\text{Accuracy} = \frac{\text{Number of correct predictions}}{\text{Total number of predictions}} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

$$\text{Precision} = \frac{\text{True positive}}{\text{Actual results}} \text{ or } \frac{TP}{TP + FP} \quad (4)$$

$$\text{Recall} = \frac{\text{True positive}}{\text{Predicted results}} \text{ or } \frac{TP}{TP + FN} \quad (5)$$

Where TP is true positive, TN is true negative, FP is false positive, and FN is false negative. The hyperparameters of the model were optimized using the grid search method. Among the selected candidate parameters, the hyperparameters with the smallest error in the test set were selected as the optimal hyperparameters using loop traversal. All methods were implemented in Python's Sklearn library.

4. Results

The research findings are structured into three main sections. First, a comparative analysis of the feature extraction performance of the two methods is presented. Subsequently, the predictions, analyses, and classifications that were performed based on the behavior features extracted by the two methods utilizing the supervised and unsupervised learning methods are discussed in sequence.

4.1. Analysis of behavioral sequence feature extraction

The analysis of the feature extraction performance for key behaviors and inter-behavior relationships in response action sequences based on the TF-IDF and Word2vec methods is presented in the following subsections.

4.1.1. Behavior sequence feature extraction based on TF-IDF

The extraction of key behaviors by the TF-IDF method was validated through feature statistics and logistic model establishment. Initially, uninformative variables were removed from the response behaviors. Specifically, in frequency analysis, 'start' and 'end' variables were excluded, while in TF-IDF analysis, 'end' actions completely correlated with the 'start' variable were eliminated. Subsequently, logistic model regression was conducted separately based on TF-IDF and frequency statistics. The coefficients of the logistic model were output and sorted according to absolute values, as shown in Table 4 and Fig. 6.

The results reveal that, compared to the frequency logistic model, the TF-IDF logistic model exhibited the highest absolute coefficient value for the 'start' variable, indicating a negative correlation with the probability of a correct answer. This suggests that longer sequences associated with the 'start' variable corresponded to higher probabilities of correct answers. The absolute coefficient value of the 'reset' variable in the TF-IDF logistic model was relatively low. About 70 % of respondents performed the 'reset' behavior, with an IDF of 0.36, indicating its common occurrence. TF-IDF effectively suppressed the impact of this action variable on the model.

The Varying One Thing at A Time (VOTAT) strategy is prominent in the feature extraction process. VOTAT, which is widely acknowledged in the educational and psychological assessment domains, is pivotal for assessing effective answering behavior and outcomes of students [52,56]. It refers to the practice of VOTAT during problem-solving actions to explore the relationship between different variables or solution paths more clearly. In the "climate control" item of PISA 2012, the VOTAT strategy is manifested in two manners: for individual actions, students adjusted only one slider while maintaining the other two at their initial positions (e.g., "1_0_0", "0_1_0"), and for action sequences, students could fix the positions of two sliders while adjusting only one slider (e.g., "0_2_0" and "0_1_0") to determine the effect of the current slider on the temperature or humidity.

In the key response actions extracted by TF-IDF (Table 4), apart from the "start" and "reset" variables, the top-ranking coefficients by absolute value were associated with the VOTAT series of behaviors, such as '1_0_0' and '0_1_0', which control other sliders to their initial state while changing only one slider. Conversely, in the frequency statistical logistic model, among the top 10 coefficients by absolute value, only two were VOTAT behaviors, namely '0_2_0' and '0_1_0'. Logistic regression analysis demonstrates the effectiveness of TF-IDF in identifying key actions compared to traditional frequency statistics.

Table 4
Comparative logistics coefficients (sorted by absolute value).

Frequency statistical variable	Coefficient	TF-IDF variable	Coefficient
2_1_-2	−0.33	start	−4.61
reset	0.27	0_2_0	2.88
−1_2_0	−0.25	0_1_0	2.75
0_2_0	0.23	0_0_2	2.26
0_1_0	0.22	0_0_1	2.13
−1_0_-2	−0.21	reset	2.04
0_-1_1	−0.19	0_-2_0	1.81
−2_-1_2	−0.17	0_0_-2	1.75
2_-1_-2	−0.16	0_-1_0	1.62
2_2_0	−0.16	2_0_0	1.56

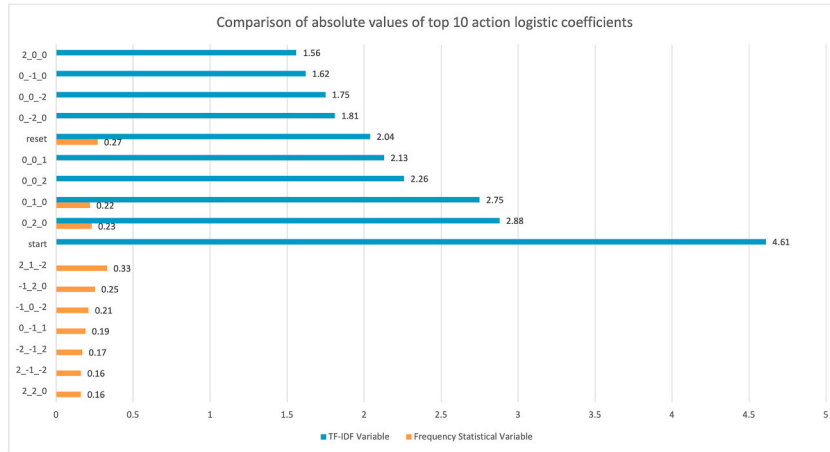


Fig. 6. Comparison of absolute values of top 10 action logistic coefficients.

4.1.2. Behavior sequence feature extraction Based on Word2vec

Based on the Word2vec method, a 100-dimensional behavior vector was generated for all behaviors, and the average of the behavior vectors was used as the numerical representation of the sequence to establish a logistic model. In general, logistic regression based on Word2vec exhibited superior performance over TF-IDF: $F1_{\text{Word2vec}} = 0.8224 > F1_{\text{TF-IDF}} = 0.8169$, $\text{Accuracy}_{\text{Word2vec}} = 0.8017 > \text{Accuracy}_{\text{TF-IDF}} = 0.7993$ and $\text{Recall}_{\text{Word2vec}} = 0.8360 > \text{Recall}_{\text{TF-IDF}} = 0.8149$.

The cosine similarity of each VOTAT behavior vector generated using Word2vec is shown in Table 5. The results show that “2_0_0” has the highest similarity with “-2_0_0”, “-2_0_0” has the highest similarity with “-1_0_0”, “0_1_0” has the highest similarity with “1_0_0”, and “2_0_0” is most similar to “0_2_0”. Based on behavior vector analysis, it was determined that the VOTAT behavior with the highest similarity for each given VOTAT behavior was also the VOTAT behavior.

The behavior vector generated by Word2vec was further reduced in 2 dimensions based on T-SNE, and the scatter plot is shown in Fig. 7. The distance between the VOTAT series behavior vectors is closer, and there is a clear linear division between them and the other behavior vectors. Combined with the results shown in Table 5, the VOTAT behavior with the highest similarity to each VOTAT behavior is also the VOTAT behavior. This indicated that Word2vec was successful in capturing latent relationships between behaviors, i.e., capturing a range of potential VOTAT behaviors.

4.2. Prediction and analysis of response performance using supervised learning models

Based on the behavior features extracted using these two text methods, the results of predicting the response performance and analyzing correct response behaviors and strategies using the supervised learning models are presented in the following subsections.

4.2.1. Predicting answering performance based on supervised learning model

Based on Word2vec and TF-IDF, KNN, SVM, and random forest were utilized. A total of 6 supervised learning models were used. The optimal hyperparameters of the models were searched using the grid search method based on 5-fold cross-validation. The F1-score, accuracy, precision, and recall of the models are shown in Table 6 and Fig. 8.

In the supervised learning model analysis, the combination of TF-IDF with Random Forest exhibited the best performance,

Table 5
Cosine similarity of VOTAT behavior vectors.

VOTAT behavior	Most similar behavior to left behavior
-2_0_0	2_0_0 (0.5786)
-1_0_0	-2_0_0 (0.5198)
1_0_0	0_1_0 (0.6745)
2_0_0	0_2_0 (0.7297)
0_-2_0	0_0_-2 (0.5836)
0_-1_0	0_0_-1 (0.5638)
0_1_0	0_0_1 (0.7735)
0_2_0	0_0_2 (0.7900)
0_0_-2	0_0_2 (0.6260)
0_0_-1	0_-1_0 (0.5639)
0_0_1	0_1_0 (0.7735)
0_0_2	0_2_0 (0.7900)

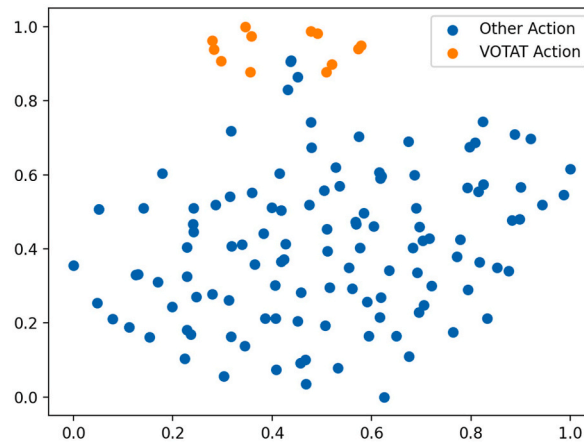


Fig. 7. T-SNE 2D scatter plot of Word2vec behavior vectors.

Table 6

Supervised machine learning analysis using TF-IDF and Word2vec feature extraction.

Indicators	Indicators		Indicators		Indicators		Indicators	
	TF-IDF	Word2vec	TF-IDF	Word2vec	TF-IDF	Word2vec	TF-IDF	Word2vec
F1-score	0.8169	0.8224	0.8008	0.8187	0.8160	0.8312	0.8419	0.8221
Accuracy	0.7993	0.8017	0.7814	0.7948	0.7987	0.8091	0.8231	0.8020
Precision	0.8189	0.8092	0.8019	0.7956	0.8194	0.8082	0.8271	0.8116
Recall	0.8149	0.8360	0.7997	0.8431	0.8127	0.8556	0.8572	0.8328

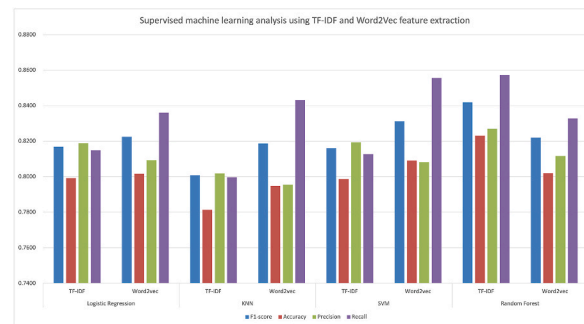


Fig. 8. Supervised machine learning analysis using TF-IDF and Word2vec feature extraction.

outperforming models based on Word2vec feature extraction and others in terms of the parameters : $RF_F1_{TF-IDF} = 0.8419 > RF_F1_{Word2vec} = 0.8221$, $RF_Precision_{TF-IDF} = 0.8271 > RF_Precision_{Word2vec} = 0.8116$, $RF_Accuracy_{TF-IDF} = 0.8231 > RF_Accuracy_{Word2vec} = 0.8020$, and $RF_Recall_{TF-IDF} = 0.8572 > RF_Recall_{Word2vec} = 0.8328$. Overall, the second-best performing model was the SVM based on Word2vec. Word2vec performed better than TF-IDF using the logistic regression, KNN, and SVM models, except in the precision. In general, both KNN models based on these two methods performed worse than the baseline (logistics) model. See Table 6 and Fig. 8 for details.

4.2.2. Correct answer-behavior and strategy analysis

The behaviors and strategies required to provide correct answers were analyzed using the feature importance of random forest based on TF-IDF. The feature importance of the random forest model is shown in Fig. 9. The results show that the most important instances are “0_2_0”, “0_0_2”, “0_1_0”, “0_0_1”, “2_0_0” and “1_0_0”. Except for “reset”, “start” and “0_0_0”, the behaviors in Fig. 9 are all VOTAT behaviors and include all VOTAT behaviors (13 behaviors in total). This indicates that among the 128 unique behaviors, these 13 VOTAT behaviors, “reset”, “start”, and “0_0_0” have the greatest impact on the random forest model. Thus, these behaviors help the random forest model to accurately predict response performance.

In addition, TF-IDF encoded “start” had a high impact on correct answers. The sequence length had an impact on the correct answer, and a longer sequence corresponded to a higher probability of obtaining a correct answer. “Reset” also had an impact on the

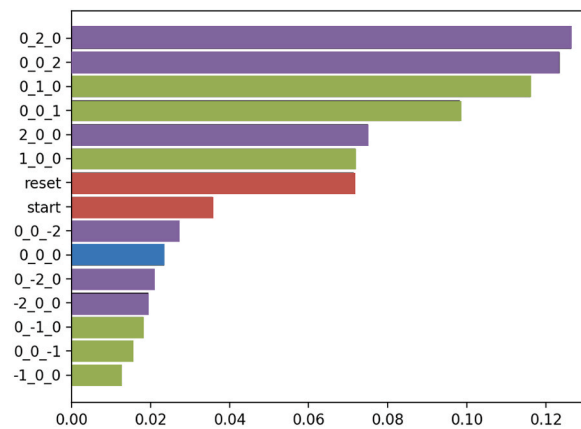


Fig. 9. Importance of random forest features.

correct answers. When the number of “resets” increased, the probability of correct answers also increased. It was determined that “0_0_0” played a role in correct answering. Further investigation revealed that there were many datasets including [“start”, “0_0_0”, “0_0_0”, “0_0_0”, “reset”, “0_0_0”, “0_0_0”, “0_0_0”, “end”] as response sequences. Thus, some respondents who frequently performed or only performed the “0_0_0” behavior lacked the interactive exploration process necessary for correct answers and could not answer the question correctly.

4.3. Cluster analysis of answer-behavior using K-means

Based on the behavior features extracted using the two text methods, the results of the unsupervised clustering analysis are presented in the following subsections.

4.3.1. K-means sequence clustering based on TF-IDF

The k-means model was established using TF-IDF. Based on the elbow principle, the optimal K value was selected as 3, and the descriptive statistics of k-means clustering are shown in Table 7 and Fig. 10. The Z test of proportions shows that the correct rate of cluster A is significantly greater than the other two clusters (cluster A vs cluster B: $Z = 46.37$, $p < 0.001$; cluster A vs cluster C: $Z = 40.27$, $p < 0.001$). In this context, “correct rate” refers to the proportion of students in a given group who answered the question correctly, whereas “behavioral patterns” refer to the identified key response behaviors and strategies (such as VOTAT) and their corresponding performance, as well as their relationship with the correct rate.

The descriptive statistics of the behavior frequency of each cluster are shown in Table 8. Cluster A has the highest average frequency of the VOTAT series behaviors, and 92 % of respondents exhibited VOTAT behaviors at least twice. In cluster B, the average frequency of the other behaviors was the highest, and only 59 % of respondents exhibited the VOTAT series behavior at least twice. The “0_0_0” behavior had the highest mean frequency in cluster C, and 83 % of respondents exhibited the “0_0_0” behavior at least twice.

The characteristics of the answer-behavior sequences in the three clusters were further analyzed, and the sample behavior sequences of each cluster were selected for observation, as shown in Table 9. The response behavior sequence of cluster A contains more VOTAT series behaviors; the behavior sequence of cluster B lacks the VOTAT series of the behaviors that are necessary for correct answers, and its behaviors are scattered and disordered. The behavior sequence of cluster C contains frequent “0_0_0” behavior.

In summary, the k-means algorithm based on TF-IDF can cluster respondents based on their behavior sequence. There were differences in the correct rate between cluster A and clusters B and C. In addition, different answer-behavior patterns affected correct answering in the sequences of the three clusters.

4.3.2. K-means sequence clustering Based on Word2vec

A k-means model was built based on Word2vec. Using the elbow principle, the optimal K value was 3. The descriptive statistics of the population are shown in Table 10. The Z test of the proportions shows that the correct rate of cluster A is significantly higher than that of the other two clusters (cluster A vs cluster B: $Z = 45.16$, $p < 0.001$; cluster A vs cluster C: $Z = 61.59$, $p < 0.001$).

The descriptive statistics of the behavior frequency of each cluster are shown in Table 11 and Fig. 11. In clusters A, B, and C, the

Table 7

TF-IDF-based k-means clustering: descriptive statistics.

Cluster	Respondent count	Correct answer count	Correct answer percentage
Cluster A	5833	4713	81 %
Cluster B	8411	3520	42 %
Cluster C	2519	896	36 %

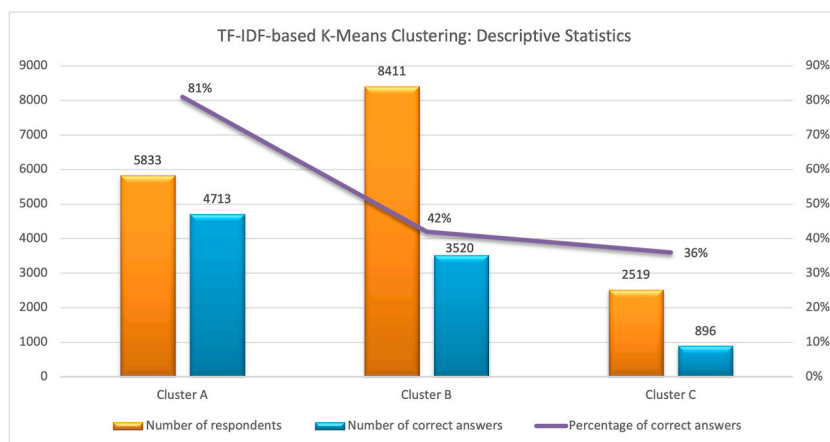


Fig. 10. TF-IDF-based k-means clustering: descriptive statistics.

Table 8

Descriptive statistics of the behavior frequency of each cluster (TF-IDF).

Cluster	Behavior	Mean	Std	Min	25 %	50 %	75 %	Max	Ratio
Cluster A	VOTAT	7.91	8.29	0	4	6	9	96	92 %
	reset	4.60	2.90	0	3	4	6	33	94 %
	0_0_0	0.54	1.13	0	0	0	1	21	12 %
	others	3.14	2.02	2	0	2	4	26	100 %
Cluster B	VOTAT	7.19	12.73	0	0	3	8	189	59 %
	reset	1.42	2.18	0	0	1	2	32	32 %
	0_0_0	1.14	2.92	0	0	0	1	52	18 %
	others	11.33	13.58	2	3	7	13	176	100 %
Cluster C	VOTAT	3.66	7.84	0	0	0	4	105	41 %
	reset	1.65	2.31	0	0	1	2	28	36 %
	0_0_0	7.65	7.66	1	2	5	11	72	83 %
	others	3.65	4.58	2	2	2	3	102	100 %

Remarks: VOTAT means 12 VOTAT behaviors; others refers to 114 behaviors except for VOTAT, “reset,” and “0_0_0”; ratio refers to the proportion of respondents who exhibited the behavior at least twice in the group. If the ratio is 50 %, this means that half of the respondents in the group exhibited the behavior at least twice.

Table 9

Sequence of response behaviors of samples in each cluster (TF-IDF).

Cluster coding	Behavior sequence
Cluster A	[start, reset, -1_0_0, -1_-1_0, -1_-1_-1, -1_0_0, -1_0_0, reset, 2_0_0, reset, 0_2_0, reset, 0_0_2, reset, 0_1_0, reset, 0_-1_0, reset, -1_0_0, reset, end]
	[start, 1_1_1, reset, 0_0_1, reset, 0_1_0, reset, 1_0_0, end]
	[start, 1_0_0, 2_0_0, -1_0_0, -2_0_0, -1_0_0, 0_0_0, reset, 0_1_0, 0_2_0, 0_-2_0, 0_2_0, 0_-2_0, reset, 0_0_2, 0_0_-2, 0_0_-1, 0_0_2, reset, end]
	[start, -1_0_0, reset, 1_0_0, reset, 2_0_0, reset, -2_0_0, reset, 0_1_0, reset, 0_2_0, reset, 0_0_1, reset, 0_0_2, reset, 0_0_-2, reset, end]
Cluster B	[start, 0_0_0, 1_2_-2, 2_2_2, 2_2_2, 2_2_2, 2_2_2, 2_2_-2, 2_2_-2, 2_2_-2, -2_-2_-2, -2_-2_-2, -2_-2_-2, -2_-2_-2, -2_-2_-2, -2_-2_0, -2_-2_0, -2_-2_0, -2_0_1, -2_0_1, -2_0_1, -2_0_1, 0_0_1, 0_0_1, 0_0_1, 0_0_1, 0_0_1, 0_0_1, 0_0_1, 0_0_1, 0_0_1, 0_0_1, 0_0_1, end]
	[start, -2_2_-2, -2_-2_-2, -1_-1_-1, 2_0_0, 0_2_0, 0_0_2, end]
	[start, reset, reset, 1_-1_1, -2_-2_-2, reset, -2_-2_-2, -1_-1_-1, 0_0_0, 1_1_1, 2_2_2, end]
	[start, 1_-1_1, 1_0_1, 1_-1_1, 1_0_1, 1_0_1, reset, 1_0_1, 2_1_2, 2_0_2, reset, 2_0_2, end]
Cluster C	[start, reset, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, reset, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, end]
	[start, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, end]
	[start, 0_0_0, 0_0_0, 0_0_0, reset, 0_0_0, 0_0_0, 0_0_0, end]
	[start, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, 0_0_0, end]

behaviors with the highest frequency mean are the VOTAT series behaviors, “0_0_0” and other behaviors. The proportion of respondents who exhibited VOTAT behaviors at least twice in clusters A, B, and C are 94 %, 29 % and 37 %, respectively.

In summary, the k-means algorithm based on Word2vec can cluster respondents based on their behavior sequence. There was a significant difference in the correct rate between cluster A and clusters B and C, and the answering sequences of the three clusters had different answer-behavior patterns.

Table 10
Word2vec-based k-means clustering: descriptive statistics.

Cluster	Respondent count	Correct answer count	Correct answer percentage
Cluster A	9245	7175	78 %
Cluster B	2115	569	27 %
Cluster C	5403	1385	26 %

Table 11
Descriptive statistics of behavior frequency of each cluster (Word2vec).

Cluster	Behavior	Mean	Std	Min	25 %	50 %	75 %	Max	Ratio
Cluster A	VOTAT	10.45	12.60	0	4	6	12	189	94 %
	reset	3.54	3.05	0	1	3	5	33	74 %
	0_0_0	0.99	2.43	0	0	0	1	61	19 %
	others	4.16	4.68	2	2	3	4	102	100 %
Cluster B	VOTAT	2.44	6.20	0	0	0	3	87	29 %
	reset	1.52	2.24	0	0	1	2	21	32 %
	0_0_0	8.24	7.93	1	2	6	12	72	81 %
	others	3.81	4.72	2	2	2	3	69	100 %
Cluster C	VOTAT	2.59	5.15	0	0	1	3	65	37 %
	reset	1.29	2.07	0	0	1	2	25	28 %
	0_0_0	0.99	2.75	0	0	0	1	50	15 %
	others	14.11	15.40	3	5	9	17	176	100 %

Remarks: VOTAT means 13 VOTAT behaviors, others refer to 114 behaviors except for VOTAT, “reset” and “0_0_0”; Ratio refers to the proportion of respondents who exhibited the behavior at least twice in the group. If the ratio is 50 %, which means that half of the respondents in the group exhibited the behavior at least twice.

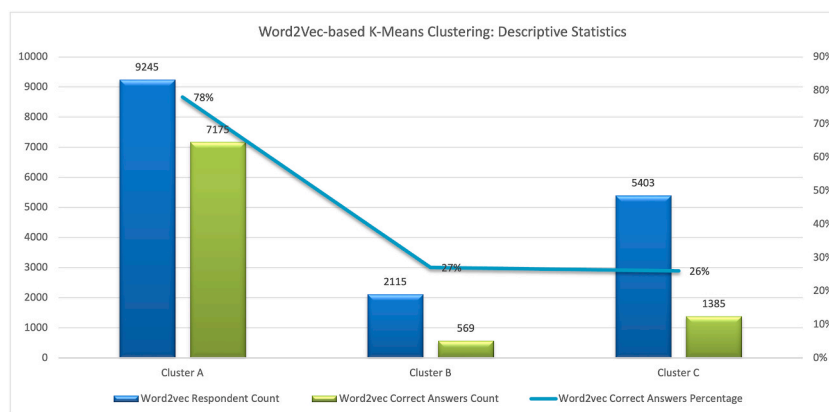


Fig. 11. Word2vec-based k-means clustering: descriptive statistics.

4.3.3. Comparison of K-means sequence clustering

Comparing the k-means classification results of the Word2vec and TF-IDF methods, the k-means algorithm of Word2vec demonstrates superior group separation ability (accuracy: 78 % vs 27 % vs 26 %) compared to the k-means algorithm of TF-IDF (accuracy: 81 % vs 42 % vs 36 %). While both algorithms produce high-accuracy clusters with similar response accuracy, Word2vec's k-means algorithm outperforms TF-IDF's k-means algorithm in segregating low-accuracy groups.

Table 12
Comparison of VOTAT strategy ratios based on k-means clustering.

	Cluster	VOTAT personnel ratio
K-means algorithm based on TF-IDF	A (correct rate: 81 %)	92 %
	B (correct rate: 42 %)	59 %
	C (correct rate: 36 %)	41 %
K-means algorithm based on Word2vec	A (correct rate: 78 %)	94 %
	B (correct rate: 27 %)	29 %
	C (correct rate: 26 %)	37 %

Remarks: VOTAT Personnel Ratio refers to the proportion of respondents who have implemented the VOTAT strategy in the cluster.

To further explore the reasons for this difference, the use of the VOTAT strategy [56] of each group in the TF-IDF-based k-means algorithm and the Word2vec-based k-means algorithm was analyzed. The results are shown in Table 12. The VOTAT strategy in this study involved the top, middle, and bottom three sliders, each of which had at least one operation in the zero state of the other two sliders. When this strategy was utilized, the top slider and the middle slider states were set as the initial state and the bottom slider state was changed at least once, for example, “0_0_1”. In addition, the top slider and bottom slider states were set as the initial state, and the middle slider state was changed at least once, for example, “0_1_0”. Finally, the middle slider and the bottom slider states were set as the initial state and the top slider state was changed at least once, for example, “1_0_0”.

The clustering results based on Word2vec and K-means exhibited lower frequencies of VOTAT usage for classes B and C with lower accuracy (Fig. 12) compared to the TF-IDF method. This suggests that the Word2vec-based k-means algorithm is better suited for identifying and distinguishing response sequences based on the VOTAT strategy than TF-IDF.

5. Discussion

Building upon the research results outlined above, separate discussions are presented for the feature extraction performance of the two methods, as well as the outcomes of the supervised prediction analysis and unsupervised clustering.

5.1. Effective extraction of behavioral sequence features based on TF-IDF and Word2vec

The analysis of the key behaviors extracted based on TF-IDF demonstrates superiority over the logistic model. The essence of TF-IDF is the utilization of the IDF as a weight to calculate the term frequency by weight. IDF assumes that the importance of a word relative to a document is inversely proportional to its frequency of appearance in all the documents, which is intuitive. TF-IDF can identify keywords that are closely related to the topic, distinguishing them from non-keywords [38]. In addition to the sequence length information contained in the TF-IDF, a possible reason for the observed results is that the TF-IDF method could reduce the effect of the commonly used behavior (reset) [37], and can also highlight the key behaviors that reflect answering strategies and contribute to correct answer, such as the VOTAT series behaviors [56]. Compared to frequency variables, the advantage of TF-IDF attenuation of commonly used behaviors with low discrimination was validated.

Word2vec outperformed the frequency statistics feature in the logistic model, as evidenced by the similarity of the VOTAT behaviors (Table 5) and the 2D scatterplot of the behavior vectors (Fig. 7). According to the adaptation environment of Word2vec, it captures potential relationships between actions and identifies a range of potential VOTAT actions. These latent behaviors can be captured using Word2vec because of the underlying “context” relationship between the VOTAT series of behaviors required for correct answers. Specifically, to answer correctly, it is necessary to perform several VOTAT series behaviors. A single VOTAT series action cannot result in a correct answer. VOTAT series behaviors often appear together, rather than individually in a behavior sequence, which is consistent with Word2vec’s theory of focusing on context and establishes the feasibility of the theoretical logic [59].

Compared to TF-IDF, Word2vec is more sensitive to policy behavior in a sequence of behaviors. Although TF-IDF is an optimization method for frequency statistics, it still has limitations in extracting position information, context information, and the distribution characteristics of some words. These attributes are particularly important in text recognition and cannot be ignored. In the field of text feature extraction, much of the work is focused on the construction of distributed low-dimensional dense word vectors, i.e., word vector representations [67], which are based on the assumption that contextually similar words have similar semantics [68]. This method maps words in a text into a fixed-length word vector via a training process, and all vectors form a word vector space. Each vector can be regarded as a point in space, and the distance between words with similar semantics in that space is closer [69]. Word2vec is more contextual, which may also explain why it is more sensitive to strategic actions in a sequence of behaviors.

In summary, TF-IDF and Word2vec successfully extracted the latent information in the sequence of question-solving answer-behaviors. First, TF-IDF and Word2vec outperformed the frequency statistical feature in the logistic model. Secondly, compared with the frequency statistical features, the advantage of TF-IDF was verified, and the influence of the common behavior “reset” on the model decreased. The most discriminative VOTAT series behaviors were highlighted. Thirdly, Word2vec successfully captured the potential relationship between behaviors. In this series, behavior vectors are closer in the vector space, and there is a clear linear division compared to other behaviors. Therefore, the first research question of this study was affirmatively answered.

5.2. Prediction of different answering performances by supervised learning model

To answer research question 2, supervised machine learning methods (KNN, SVM, and random forest) were applied to analyze the behavior features extracted by TF-IDF and Word2vec. The results reveal that the random forest model based on TF-IDF outperformed the others overall and surpassed Word2vec. SVM performed better on the Word2vec-derived behavior features compared to TF-IDF. This suggests that random forest with TF-IDF and SVM with Word2vec can effectively predict the response behaviors of respondents.

The potential reasons for the superior performance of the TF-IDF and random forest models are as follows: (1) TF-IDF characteristics: TF-IDF effectively captures key terms by providing importance scores based on the word frequency and distribution in the document collection, filtering out common words. (2) Adaptability of random forest: Random forest enhances the model performance by constructing multiple decision trees. Decision tree models excel with sparse data and the high-dimensional sparse matrix of TF-IDF aligns well with this feature, enabling effective feature selection. In contrast, Word2vec generates dense vectors that capture semantic relationships but may not be as directly useable in decision tree models as the sparse features of TF-IDF. (3) Direct correlation of features: The TF-IDF features directly correlate with the decision criteria used in random forest, facilitating effective differentiation

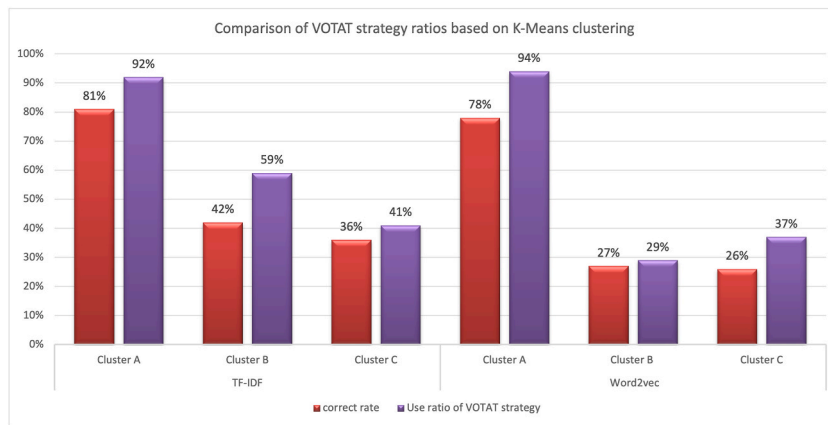


Fig. 12. Comparison of VOTAT strategy ratios based on k-means clustering.

between data classes. These features, which reflect the word frequency and distribution, align well with the ability of random forest to handle such features. (4) Model randomness and feature selection: The randomness of random forest entails random feature selection during decision tree construction, with features that offer clearer data splits preferred. This gives TF-IDF features an edge over Word2vec features. In summary, the high-dimensional sparsity, relevance to the task, and compatibility with random forest of TF-IDF may contribute to its superior performance over Word2vec. However, when selecting TF-IDF or Word2vec, the specific requirements and characteristics of the task at hand should be considered.

The superior performance of the Word2vec method over TF-IDF in the SVM model analysis can be attributed to the following factors: (1) Capturing of semantic information: Word2vec represents words as dense vectors, capturing the semantic relationships between words. In contrast, TF-IDF generates sparse high-dimensional vectors without capturing the semantic similarity. (2) Dimensionality and sparsity of feature vectors: Word2vec generates dense feature vectors with smaller dimensions, aiding SVM in effectively learning the data distribution, whereas the high-dimensional sparse vectors of TF-IDF may pose challenges for SVM training. (3) Ability to recognize complex patterns: SVM with Word2vec-encoded features can identify complex data distribution patterns more effectively than TF-IDF owing to its capability to determine optimal decision boundaries in high-dimensional space. SVM is a hyperplane-based classifier, which is better suited to the Word2vec method that maps behaviors to a vector space [70,71]. (4) Task demand for semantic information: Word2vec-based models excel in tasks requiring an understanding of the semantic relationships between words, such as sentiment analysis or text similarity matching, in which TF-IDF may fall short owing to its focus on statistical word occurrences. In summary, Word2vec-based SVM outperforms TF-IDF-based models in certain scenarios owing to its rich semantic information and efficient data processing. However, the choice of method should depend on the task requirements and data characteristics.

The F1-score, accuracy, and recall metrics for the Word2vec features outperformed those of the TF-IDF features using the logistic regression, KNN, and SVM models, except in the precision. Possible reasons for this phenomenon include the following: (1) Positive class recognition: Precision measures the ratio of actual positive samples among those predicted as positive. Capturing more true positives may increase the false positives, lowering the precision. TF-IDF tends to predict positive samples conservatively, yielding higher precision than Word2vec. Despite the lower recall for positive samples, TF-IDF ensures that the predicted positives are indeed positive, leading to higher precision. (2) Semantic richness and overfitting: Word2vec captures semantic relationships, enhancing the model accuracy but risking overfitting with complex models or limited data. Overfitting may lower the precision on unseen data. The simpler features of TF-IDF help to prevent overreliance on specific semantics, leading to higher precision. (3) Model preference and decision boundaries: The logistic regression, KNN, and SVM models tend to learn complex decision boundaries with Word2vec features, enhancing the recall and accuracy. However, this complexity may lead to more aggressive classification near the boundary, reducing the precision. TF-IDF features, which are high-dimensional and sparse, result in simpler or more conservative decision boundaries, improving the precision at the expense of the recall. (4) Dataset imbalance: In datasets with imbalanced labels (i.e., a large difference in the number of positive and negative samples), models may favor predicting the larger class. Word2vec features, which have rich context, may sacrifice precision for higher recall. Conversely, TF-IDF features, which are more conservative, tend to maintain better precision when handling imbalanced data. In conclusion, various performance metrics must be considered during feature selection and model training, while balancing the importance of the precision and recall.

In addition, this study underscores the efficacy of TF-IDF and Word2vec text methods in extracting features for process-based answering performance. By delving into VOTAT strategies and typical answering behaviors [56,72,73], the research unveils crucial connections between these behaviors and the answer accuracy, offering valuable insights for enhancing educational assessments and understanding student answering behaviors. Furthermore, the study revealed correlations between the action sequence length and reset frequency with the probability of correct answers [72], as well as the "0_0_0" resetting behavior and low correct answer rates [73]. These findings highlight the importance of considering the multidimensionality of data, harnessing the potential of behavioral data, ensuring model fairness and minimizing biases, leveraging models for practical applications, and continuous iteration and

improvement when designing and implementing machine learning solutions.

In conclusion, Supervised learning methods, including KNN, SVM, and Random Forest, were employed to analyze behavior features extracted by TF-IDF and Word2vec. Random Forest with TF-IDF exhibited superior performance, while SVM performed better with Word2vec-derived features. TF-IDF's strength lies in key term capture and compatibility with decision trees, while Word2vec excels in semantic information capture and efficient data processing, particularly with SVM. Despite Word2vec's advantages, TF-IDF remains competitive due to its sparsity and task relevance. The study offers insights for educational assessments and student behavior understanding.

5.3. Cluster analysis

Using TF-IDF and Word2vec, the k-means algorithm analyzed the clusters of respondents based on their behavior sequences. Three distinct clusters were identified, with one cluster showing a significantly higher correct response rate compared to the other two clusters. This cluster exhibited more VOTAT strategies, aligning with previous findings that linked behavioral strategies to problem-solving processes [73]. The other two clusters had lower accuracy rates. One cluster displayed scattered and disordered behavior sequences, indicating a lack of VOTAT strategy ability. The second cluster frequently or exclusively exhibited a "0_0_0" behavior, suggesting a lack of clarity in answering concepts and a failure to extract effective answering strategies from the behavior sequence. These results confirmed the clustering effect observed in previous studies [53,73] and validated the usefulness of TF-IDF and Word2vec in extracting behavioral features.

The reasons for the relatively good clustering performance that was achieved through the analysis of the characteristics of Word2vec, TF-IDF, and the k-means algorithm are explained as follows:

Word2vec offers semantic richness by mapping words into a vector space based on large textual datasets, enabling it to capture semantic similarities between words. In addition, Word2vec is context sensitive and considers the surrounding context of words during training, thereby accurately capturing the word semantics, which is crucial for understanding complex response behaviors.

TF-IDF quantifies importance by evaluating the frequency of words in a specific document relative to their distribution across all documents. This method identifies keywords that are more discriminative for specific response content, which helps to reveal the characteristics of different response behaviors. In addition, TF-IDF filters out common vocabulary by reducing the weight of words that appear frequently across all documents. This helps to reduce interference from common words in classification, thereby allowing the model to focus more on distinguishing unique vocabulary associated with different response behaviors.

As an unsupervised clustering technique, k-means categorizes data by minimizing the intra-cluster distances and maximizing the inter-cluster distances. Two key aspects are notable when using features extracted from Word2vec and TF-IDF for k-means clustering. First, it facilitates high-dimensional feature representation: Word2vec and TF-IDF generate feature vectors that map response behaviors to high-dimensional space, offering rich information for k-means to discern and differentiate diverse response behavior patterns. Second, it underscores substantial category distinctions: Word2vec captures profound semantic relationships, whereas TF-IDF accentuates the significance of pivotal words. The amalgamation of features extracted by these methodologies comprehensively reflects the attributes of response behaviors. Hence, k-means clustering leveraging these features adeptly discriminates various facets of response behaviors, yielding noticeable disparities in classification accuracy.

In summary, Word2vec and TF-IDF extract textual features from both semantic and importance dimensions, furnishing k-means clustering with rich and effective information. This enables the algorithm to distinguish various problem-solving response behaviors in high-dimensional space proficiently, thereby manifesting significant disparities in classification accuracy.

6. Conclusions

Based on the PISA 2012 climate control problem-solving task, TF-IDF and Word2vec were used to analyze problem-solving behavior sequences. Supervised learning algorithms were used to predict performance, analyze behavior, and identify strategies for correct answers. Cluster analysis was also employed to group respondents. The main findings are.

- 1) TF-IDF and Word2vec, two text feature extraction methods, effectively extracted latent information from answer-behavior sequences during problem-solving. TF-IDF effectively identified key answering actions, confirming its theoretical advantage in suppressing commonly used behaviors with low discrimination, and Word2vec successfully captured the potential relationship between behaviors and identified a series of potential VOTAT behaviors.
- 2) TF-IDF and Word2vec were used to accurately predict the answering performance of respondents using supervised learning algorithms. The random forest model based on TF-IDF performed best, followed by the Word2vec SVM model. Word2vec-based models outperformed TF-IDF-based models in F1-score, Accuracy, and Recall across Logistic Regression, KNN, and SVM (except for Precision). The VOTAT behavior strategy detected by the models was essential for correct answers.
- 3) Unsupervised machine learning with k-means clustering effectively groups behavior features that are identified by TF-IDF and Word2vec. The different clusters showed distinct response behavior patterns, accuracy rates, and VOTAT usage proportions. Higher accuracy clusters exhibited increased VOTAT usage. Word2vec outperformed TF-IDF in recognizing VOTAT actions with sequential properties.

Contributions

The exploration of problem-solving behavior sequences using TF-IDF and Word2vec can be summarized as follows: First, this study successfully employed cross-disciplinary techniques by using TF-IDF and Word2vec for feature extraction from response actions and sequences. These text analysis methods captured essential information and deep semantic relationships, providing a rich feature representation that is not commonly found in traditional educational assessment analysis. Second, this approach serves as an effective supplement for acquiring fine-grained assessment evidence. The research combines supervised and unsupervised machine learning. This not only allows for in-depth analysis of effective response strategies, but also validates the effectiveness of the text method for feature extraction and the response assessment model. Third, it fosters personalized assessment and learning, offering the potential for finely tailored educational interventions by discerning diverse student response patterns with granularity. This capability can facilitate the design of targeted teaching strategies and interventions. Fourth, novel tools and perspectives are offered for educational assessment. This study showcases the power of advanced text analysis and machine learning techniques in unearthing valuable assessment evidence from large-scale educational data. It provides practical support for innovative assessment approaches and offers a fresh perspective on assessment methods and strategies.

Limitations

Although this study has delved into the cross-disciplinary feature extraction of the TF-IDF and Word2vec text methods, several limitations remain. First, considering more advanced approaches is advisable when using Word2vec for sequential representation. The averaging method employed in this study may result in the loss of information such as sequential details, potentially impacting the overall model accuracy adversely. **Future research could explore advanced methods, such as the use of smoothed inverse frequency for weighted averaging or behavioral vectors as embedding layers for constructing deep learning models. Second, exploring alternative text analysis methods beyond TF-IDF and Word2vec to enhance the contextual sensitivity is crucial.** For instance, TF-IDF may lack semantic understanding and may favor high-frequency terms, potentially overlooking the importance of less frequent terms, particularly in longer documents. In addition, Word2vec-generated word vectors may not effectively address biases that are present in the training set, and the fixed window size of the context may limit the ability of the model to capture contextual information over longer ranges. Other methods to consider may include Bidirectional Encoder Representations from Transformers, long short-term memory, Generative Pre-trained Transformer, text graph representations, sentiment analysis techniques, and topic modeling methods. Third, parameter tuning and model selection pose challenges. Determining optimal TF-IDF and Word2vec parameter configurations requires extensive experimentation, which results in research process complexity and excessive time consumption. Furthermore, the choice between supervised and unsupervised models significantly impacts the results. Finally, model generalizability is a concern. Given the reliance of this study on specific datasets, overfitting risks exist, where models perform well on certain datasets but lack generalizability to new or different types of datasets. Validating research findings on broader datasets is necessary to enhance the generalization and transferability.

In summary, applying TF-IDF and Word2vec for extracting behavioral features from problem-solving processes holds theoretical and transferable significance. In this study, key response behaviors and strategies were effectively extracted by leveraging contextual analysis, thereby overcoming previous limitations and enhancing the precision of machine learning models. This research not only advances the evidence value of process data assessment, but also provides valuable insights for the education and psychological evaluation domains. The findings of this study can be further validated and explored through methodological refinements and the use of diverse datasets.

Data availability statement

The data used in this study were sourced from the open database of the OECD official website, with the ultimate interpretive authority vested in the OECD. This database is publicly available and can be obtained from the following link: <https://www.oecd.org/pisa/data/pisa2012database-downloadabledata.htm>. The climate control dataset originated from the ProcData package in R language.

Generalization statement

The conclusions of this study have not been extended to other datasets of answer-behavior sequences in problem-solving. We anticipate validation and discussion among peers.

Ethics declaration statement

Review and/or approval by an ethics committee was not required for this study because the data were sourced from the OECD public database, and ethical considerations regarding personnel are not applicable to the research contents.

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CRedit authorship contribution statement

Jing Zhou: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis. **Zhanliang Ye:** Writing – original draft, Software, Formal analysis, Data curation. **Sheng Zhang:** Validation, Supervision, Methodology, Investigation, Conceptualization. **Zhao Geng:** Software, Formal analysis, Data curation. **Ning Han:** Writing – review & editing, Visualization, Software, Formal analysis. **Tao Yang:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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